

AUGUSTINE ATTA DEBRAH

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School of Chemistry and Biochemistry, Georgia Institute of Technology, Atlanta, GA 30332

PROFESSIONAL SUMMARY

Ph.D. Candidate in Analytical Chemistry at the Georgia Institute of Technology, specializing in microfluidic droplet systems for prebiotic chemical evolution studies. My research focuses on designing and characterizing a water-in-gas droplets generation system to investigate droplet formation dynamics to simulate early-Earth chemical processes. Supported by funding from Scialog: Signatures of Life in the Universe, a competitive initiative of the Research Corporation for Science Advancement (RCSA), my work advances the development of experimental tools and analytical frameworks for studying prebiotic chemical evolution, atmospheric processes, and planetary habitability while contributing to scientific training and interdisciplinary research.

EDUCATION

Ph.D. in Analytical Chemistry (*Expected 2027*)

Georgia Institute of Technology, Atlanta, GA

Advisor: Prof. Amanda Stockton

Research Focus: Prebiotic Chemical Evolution, Microfluidic droplet generation, Atmospheric Chemistry, Instrumentation, Origins of Life.

Certificate in Technology Innovation (TI: GER Program) (*Expected Dec. 2026*)

Scheller College of Business, Georgia Institute of Technology, Atlanta, GA

M.S. in Chemistry (Non-Thesis) (2022 – 2025)

Georgia Institute of Technology, Atlanta, GA

B.Sc. in Laboratory Technology (2012 – 2016)

University of Cape Coast, Ghana

Undergraduate Research: Designed and built an RF mobile-phone signal detector for restricted cellphone environments.

EXPERIENCE

Graduate Research Assistant

Stockton Laboratory, Georgia Institute of Technology (2022 – Present)

- Developed and validated an aqueous-in-gas microfluidic system for simulating aerosol transport processes relevant to planetary environments.
- Established droplet generation stability regimes and characterized flow-rate dependent droplet behavior, size distributions, and operating transitions.
- Investigate the transport and concentration of simple organic molecules within aerosol

microenvironments relevant to chemical evolution.

- Test amino acid condensation within microfluidic droplets with amino acids like glycine under drying conditions and UV exposure.
- Investigate ribozyme assembly through freeze-thaw cycling of RNA subunits within the microfluidic droplets.
- Conduct spectroscopic and optical characterization of microfluidic droplets for the processing of atmospheric signatures relevant to exoplanet characterization.
- Research Support: *Scialog: Chemical Machinery of the Cell, promoting innovation in chemical systems research relevant to early-Earth chemistry.*

Graduate Teaching Assistant

Georgia Institute of Technology (2022 – Present)

- Train 100+ undergraduate students in analytical instrumentation, teaching hands-on experiments in chromatographic and spectroscopic techniques (GC, GC-MS, LC, Raman, FTIR, UV-Vis, LIBS, and Flow Injection analysis).
- Guide students in sample preparation, method development, instrument operation, calibration, and data interpretation.
- Author and revise Standard Operating Procedures (SOPs), ensuring consistent and traceable laboratory practices

Laboratory Technologist & Research Assistant

University of Cape Coast, Ghana (2018 – 2022)

- Instructor on record for Analog and Digital electronics, Instrumentation, and Engineering materials laboratories for over 100 students per year.
- Designed and tested circuit systems and laboratory devices for educational and research use.
- Implemented preventive maintenance and calibration schedules to sustain instrument functionality and safety.
- Provided hands-on training in method development, instrument design, operation, and data collection for Electronics, Microbiology, and Food Science and Technology instrumentation research projects.
- Supervised undergraduate research projects in Electronics, Microbiology, and Food-Science and Technology.

Laboratory Assistant (Internship)

Ghana Manganese Company Ltd, Ghana (May, 2015 – July, 2015)

- Prepared ore samples and standard solutions for analysis.
- Performed manganese determination via redox titrimetry using potassium permanganate after acid digestion with hydrochloric and nitric acids.
- Assisted in boron analysis using spectrophotometric methods for water and

environmental samples.

- Documented analytical results, ensured data accuracy, and supported quality control procedures in both the Assay and Environmental laboratories.

PROJECTS & INITIATIVES

Founder & Program Lead – Analytical Instrumentation Training (AIT) Program

Georgia Institute of Technology (2025 – Present)

- Collaborated with faculty and students to design curriculum structure and training modules for a departmental analytical instrumentation program.
- Provide hands-on operation in the operation of AAS, GC, GC-MS, LC, Raman, FTIR, and UV-Vis instrumentation.
- Organize training sessions emphasizing data processing and quality control of analytical instruments.

UNDERGRADUATE RESEARCH MENTORSHIP

Georgia Institute of Technology (2024-Present)

- Mentor undergraduate researchers in analytical and microfluidic system development.
- Train students in experimental design, data collection, instrument operation, and laboratory safety.
- Guide scientific communication, data interpretation, and research methodology.
- Support student participation in independent research projects related to analytical instrumentation and aerosol microfluidics

Undergraduate Researchers Mentored

- **Kenneth Nguyen (Fall 2025-Present)**
Project: Environmental control chamber development for simulated Aerosol Microreactors
 - **Isra Hussain (Fall 2025-Present)**
Project: Computational Analysis of Microfluidic Droplet Formation Dynamics
 - **Manas Gangwar (Fall 2025-Spring 2026)**
Project: Microfluidic Droplet Generation and High-speed Imaging Analysis
 - **Amanda Chesney (Fall 2025)**
Project: Effects of Fluid Viscosity on Microfluidic Droplet Generation and Stability
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CONFERENCE & WORKSHOPS

- ExxonMobil-PALS Georgia Tech Program 2026, Houston, TX – *School of Chemistry and Biochemistry representative at PALS two-day workshop.*
- ACS Spring 2026, Atlanta, GA – *Poster: liquid-liquid to liquid-gas: Expanding the*

design space of microfluidic droplet generation.

- ASME Fluids Engineering Division Summer Meeting, Philadelphia, PA (2025) — *Technical/Oral Presentation: Development of a Sea-Spray Microfluidic Droplet System for Chemical Evolution Studies.*
- AbGradCon 2025, Boulder, CO — *Poster: Sea Spray Aerosol Microfluidics: Simulating Prebiotic Atmospheric Chemistry*
- Pittcon 2025, Boston, MA — *Poster: Design, Assembly, and Application of Sea Spray Aerosol Microfluidic Droplet Generator for Experimental Atmospheric Simulations.*

LEADERSHIP & SERVICE

- Graduate Student Representative, Undergraduate & Graduate Academic Programs Committee (UGAP), Georgia Tech (2025 – Present)
- President, African Graduate Students Connect — Georgia Tech (2025 – Present)
- President, Ghanaian Students Association — Georgia Tech (2023 – 2025)
- Vice President, Black Graduate Students Association — Georgia Tech (2023 – 2024)
- Graduate Senator, Graduate Student Government Association — Georgia Tech (2023 – 2024)
- Vice President, University Laboratory Technologists Association — UCC, Ghana (2019 – Present)
- President, Laboratory Technology Students Association- UCC, Ghana (2014-2015)

AWARDS & RECOGNITION

- Conference Travel Grant - Georgia Tech College of Sciences Graduate Career Connect Travel Grant (2026). *Award to be used towards transportation, hotel, conference registration, and per diem costs.*
- AbGradCon Conference Award (2025) – *Conference funding for travel, lodging, meals, and per diem expenses.*
- White & Gold Student Leadership Award (2024) - Georgia Tech Center for Student Engagement. *Leadership award for founding the Georgia Tech Ghanaian Students Association*
- Graduate Teaching Assistantship, Georgia Institute of Technology (2022 – Present) *Full funding for doctoral studies awarded through competitive selection.*

INVITED TALKS & PANELS

Panelist – Collaboration Over Competition: Strengthening Leadership Networks
Center for Student Engagement, Georgia Institute of Technology, August 2025

- Contributed as a panelist to a university-wide leadership forum focused on collaborative innovation and peer empowerment.

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- Represented graduate student leadership, sharing practical strategies for fostering interdisciplinary teamwork and mentoring across communities.

PROFESSIONAL MEMBERSHIPS

- American Chemical Society (ACS)
- Association of Official Analytical Collaboration (AOAC International)
- Society for Laboratory Automation and Screening (SLAS)
- Society of Environmental Toxicology and Chemistry (SETAC)

TECHNICAL EXPERTISE

- **Analytical Techniques:** GC-FID/ECD, GC-MS, LC, Raman, FTIR, UV-Vis, LIBS, Flow Injection Analysis, microfluidic droplet systems, instrument prototyping, method development, process optimization.
- **Software:** Python, JMP, MaxQuant, MZmine, Multisim, Biorender, ChemDraw, Microsoft Office Suite.
- **Other Skills:** Experimental design, data analysis, scientific communication, teaching and mentorship, troubleshooting analytical systems, and project management.

PUBLICATIONS

- Hou, H., Yang, S., Yang, X., Sun, W., **Debrah, A. A.**, Javeria, H., Tian, D., & Du, Z. (2025). Comprehensive profiling and development of a collision cross-section database for milk oligosaccharides via orthogonal UPLC-cyclic ion mobility-mass spectrometry system. *Food Chemistry*, 480, 143839. <https://doi.org/10.1016/j.foodchem.2025.143839>
- Hou, H., Wang, M., Yang, S., Yang, X., Sun, W., Sun, X., Guo, Q., **Debrah, A.A.**, & Du, Z. (2024). Evaluation of prebiotic glycan composition in human milk and infant formula: profile of galacto-oligosaccharides and absolute quantification of major milk oligosaccharides by UPLC-Cyclic IM-MS and UPLC-MS/MS. *Journal of Agricultural and Food Chemistry*, 72(14), 7980-7990. <https://doi.org/10.1021/acs.jafc.4c00810>
- Hou, H., Wu, Q., Li, Z., Wang, D., **Debrah, A. A.**, Zou, J., & Du, Z. (2023). A non-invasive and rapid method for discrimination of inkjet printouts by ion mobility spectrometry combined with chemometrics. *Microchemical Journal*, 194, 109250. <https://doi.org/10.1016/j.microc.2023.109250>
- Hou, H., Tang, Y., Zhao, J., **Debrah, A. A.**, Shen, Z., Li, C., & Du, Z. (2023). Authentication of organically produced cow milk by fatty acid profile combined with chemometrics: A case study in China. *Journal of Food Composition and Analysis*, 120, 105297. <https://doi.org/10.1016/j.jfca.2023.105297>

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- Wang, M., Zhang, F., **Debrah, A. A.**, Khan, J., Hou, H., Yuan, Q., & Du, Z. (2023). Selective extraction of phospholipids from human milk using glass fabric modified with zirconium-based metal organic framework. *Journal of Chromatography A*, 1710, 464435. <https://doi.org/10.1016/j.chroma.2023.464435>
 - Wang, M., Zhang, F., Tang, Y., Ali, M. M., Shen, Z., **Debrah, A. A.**, Yuan, Q., & Du, Z. (2022). Boron-doped titania for separation and purification of lactoferrin in dairy products. *Journal of Chromatography B*, 1212, 123501. <https://doi.org/10.1016/j.jchromb.2022.123501>
 - Sun, X., Zheng, X., Tang, Y., **Debrah, A. A.**, & Du, Z. (2021). Supercritical fluid extraction combined with ultrahigh performance liquid chromatography quadrupole time-of-flight mass spectrometry for determination of extractables to evaluate compatibility of drugs with rubber closures. *AAPS PharmSciTech*, 22, 1-9. <https://doi.org/10.1208/s12249-020-01907-6>
 - Tang, Y., Ali, M. M., Sun, X., **Debrah, A. A.**, Wang, M., Hou, H., Guo, Q., & Du, Z. (2021). Development of a high-throughput method for the comprehensive lipid analysis in milk using ultra-high performance supercritical fluid chromatography combined with quadrupole time-of-flight mass spectrometry. *Journal of Chromatography A*, 1658, 462606. <https://doi.org/10.1016/j.chroma.2021.462606>
 - Tang, Y., Sun, X., Hou, H., **Debrah, A. A.**, Wang, M., Zhang, F., & Du, Z. (2021). Investigation of alterations in phospholipids during the production chain of infant formulas via HILIC-QTOF-MS and multivariate data analysis. *Food Chemistry*, 364, 130414. <https://doi.org/10.1016/j.foodchem.2021.130414>
 - Ma, X., Sui, H., Sun, X., Ali, M. M., **Debrah, A. A.**, & Du, Z. (2021). A risk classification strategy for migrants of food contact material combined with three (Q) SAR tools in silico. *Journal of Hazardous Materials*, 419, 126422. <https://doi.org/10.1016/j.jhazmat.2021.126422>
 - Hu, Y., Du, Z., Sun, X., Ma, X., Song, J., Sui, H., & **Debrah, A. A.** (2021). Non-targeted analysis and risk assessment of non-volatile compounds in polyamide food contact materials. *Food Chemistry*, 345, 128625. <https://doi.org/10.1016/j.foodchem.2020.128625>
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